Class: **CAPE Pure Mathematics Unit 1**

Topic: **Past Paper June 2017 – Paper 2 – Solutions**

June 2017 – Question 1 – Solution

(a) Let p and q be two propositions.

p : It is raining.

q : John is sick.

Write EACH of the statements below in terms of p and q.

(i) It is not raining or John is sick. [1]

Symbolic form: $\sim p \vee q$

(ii) If it is raining then John is not sick. [1]

Symbolic form: $p \rightarrow \sim q$

(b) An operation $*$ is defined on the set $\{1, 2, 3, 4\}$ as shown in the following table.

$*$	**1**	**2**	**3**	**4**
1	2	4	1	3
2	4	3	2	1
3	1	2	3	4
4	3	1	4	2

(i) Prove that $*$ is commutative. [1]

Required to prove that $*$ is commutative.

$*$	**1**	**2**	**3**	**4**
1	2	4	1	3
2	4	3	2	1
3	1	2	3	4
4	3	1	4	2

The table is symmetric about the leading diagonal, hence, $*$ is commutative.

That is, $a * b = b * a$, for all a, b.

Q.E.D.

(ii) Show that the identity element of $*$ is 3. [2]

$*$	**1**	**2**	**3**	**4**
1	2	4	1	3
2	4	3	2	1
3	1	2	3	4
4	3	1	4	2

The identity element is 3.

Additionally,

$1 * 3 = 3 * 1 = 1$

$2 * 3 = 3 * 2 = 2$

$3 * 3 = 3 * 3 = 3$

$4 * 3 = 3 * 4 = 4$

$\therefore a * 3 = 3 * a = a$ for all a in $\{1, 2, 3, 4\}$.

Hence, 3 is the identity element in $*$.

Q.E.D.

(c) The polynomial $f(x) = ax^3 + 9x^2 - 11x + b$ has a factor of $(x - 2)$ and a remainder of 12 when divided by $(x + 2)$.

(i) Show that $a = 2$ and $b = -30$. [4]

$f(x) = ax^3 + 9x^2 - 11x + b$

Since $(x - 2)$ is a factor of $f(x)$, then by the Factor Theorem, $f(2) = 0$.

$f(2) = a(2)^3 + 9(2)^2 - 11(2) + b$

$0 = 8a + 36 - 22 + b$

$0 = 8a + 14 + b$

$b = -8a - 14$ → Equation 1

Since $f(x)$ has a remainder of 12 when divided by $(x + 2)$, then by the Remainder Theorem, $f(-2) = 12$.

$f(-2) = a(-2)^3 + 9(-2)^2 - 11(-2) + b$

$12 = -8a + 36 + 22 + b$

$12 = -8a + 58 + b$

$b = 8a + 12 - 58$

$b = 8a - 46$ → Equation 2

Equating Equation 1 and Equation 2 gives:

$-8a - 14 = 8a - 46$

$8a + 8a = 46 - 14$

$16a = 32$

$a = \frac{32}{16}$

$a = 2$

Substituting $a = 2$ into Equation 2 gives:

$b = 8(2) - 46$

$= 16 - 46$

$= -30$

$\therefore a = 2$ and $b = -30$

Q.E.D.

(ii) Hence, solve $ax^3 + 9x^2 - 11x + b = 0$. [9]

Since $a = 2$ and $b = -30$, we have $f(x) = 2x^3 + 9x^2 - 11x - 30$.

We are given that $(x - 2)$ is a factor of $f(x)$.

By long division,

$$
\begin{array}{r|l}
 & 2x^2 + 13x + 15 \\
\hline
x - 2 & 2x^3 + 9x^2 - 11x - 30 \\
 & 2x^3 - 4x^2 \\
\cline{2-2}
 & 13x^2 - 11x \\
 & 13x^2 - 26x \\
\cline{2-2}
 & 15x - 30 \\
 & 15x - 30 \\
\cline{2-2}
 & 0 \\
\cline{2-2}
\end{array}
$$

So, we have,

$f(x) = (x - 2)(2x^2 + 13x + 15)$

$= (x - 2)(2x + 3)(x + 5)$

When $f(x) = 0$,

$(x-2)(2x+3)(x+5) = 0$

Either $x - 2 = 0$ or $2x + 3 = 0$ or $x + 5 = 0$

$x = 2$ $x = -\frac{3}{2}$ $x = -5$

$\therefore x = -5, -\frac{3}{2}$ or 2

(d) Use mathematical induction to prove that

$8 + 16 + 24 + 32 + \cdots + 8n = 4n(n+1)$ for all $n \in \mathbb{N}$. [7]

Let $P(n)$ be the statement: $8 + 16 + 24 + 32 + \cdots + 8n = 4n(n+1)$ for all $n \in \mathbb{N}$.

Looking at $P(1)$:

L.H.S. $= 8$

R.H.S. $= 4(1)(1+1)$

$= 4(1)(2)$

$= 8$

Since L.H.S. = R.H.S., $P(1)$ is true.

Assume $P(k)$ is true.

That is, $8 + 16 + 24 + 32 + \cdots + 8k = 4k(k+1)$

Looking at $P(k+1)$.

That is, $8+16+24+32+\cdots+8(k+1)=4(k+1)(k+2)$

We need to prove $P(k+1)$.

Taking L.H.S:

$$\begin{aligned}8+16+24+32+\cdots+8(k+1) &= 8+16+24+32+\cdots+8k+8(k+1)\\ &= 4k(k+1)+8(k+1) \qquad \text{[from assumption]}\\ &= 4(k+1)(k)+4(k+1)(2)\\ &= 4(k+1)(k+2)\\ &= \text{R.H.S.}\end{aligned}$$

$\therefore$ If $P(k)$ is true, then $P(k+1)$ is also true.

Since $P(1)$ is true, and since $P(k+1)$ is true when $P(k)$ is true, then by the principle of mathematical induction, $P(n)$ is true for all n, ie.

$8+16+24+32+\cdots+8n=4n(n+1)$ for all $n\in\mathbb{N}$.

Q.E.D.

Total: 25 marks

June 2017 – Question 2 – Solution

(a)(i) Given that $a^2 + b^2 = 14ab$, prove that $\ln\left(\frac{a+b}{4}\right) = \frac{1}{2}(\ln a + \ln b)$. [5]

We have,

$$a^2 + b^2 = 14ab$$

$$a^2 + 2ab + b^2 = 14ab + 2ab$$

$$(a+b)^2 = 16ab$$

$$\frac{(a+b)^2}{16} = ab$$

$$\left(\frac{a+b}{4}\right)^2 = ab$$

Taking ln on both sides:

$$\ln\left(\frac{a+b}{4}\right)^2 = \ln ab$$

$$2\ln\left(\frac{a+b}{4}\right) = \ln a + \ln b$$

$$\ln\left(\frac{a+b}{4}\right) = \frac{1}{2}(\ln a + \ln b)$$

$$\therefore \ln\left(\frac{a+b}{4}\right) = \frac{1}{2}(\ln a + \ln b)$$

Q.E.D.

(ii) Solve the equation $2^{-x} + 3(2^x) = 4$. [6]

[Your response may be expressed in terms of logarithms.]

$$2^{-x} + 3(2^x) = 4$$

$$(\times 2^x)$$

$$2^{-x} \cdot 2^x + 3(2^{2x}) = 4(2^x)$$

$$1 + 3(2^{2x}) = 4(2^x)$$

Let $y = 2^x$.

Then, we have,

$$1 + 3y^2 = 4y$$

$$3y^2 - 4y + 1 = 0$$

$$(3y - 1)(y - 1) = 0$$

Either $3y - 1 = 0$ or $y - 1 = 0$

$y = \frac{1}{3}$ $\quad$ $y = 1$

When $y = \frac{1}{3}$,

$$2^x = \frac{1}{3}$$

$$\ln 2^x = \ln\left(\frac{1}{3}\right)$$

$$x \ln 2 = \ln\left(\frac{1}{3}\right)$$

$$x = \frac{\ln\left(\frac{1}{3}\right)}{\ln 2}$$

$$x = \frac{\ln 3^{-1}}{\ln 2}$$

$$x = -\frac{\ln 3}{\ln 2}$$

When $y = 1$,

$$2^x = 1$$

$$2^x = 2^0$$

$$x = 0$$

$\therefore x = -\frac{\ln 3}{\ln 2}$ or $x = 0$

(b) The following diagram shows the graph of the function $f(x) = \frac{3x-4}{x+4}$.

On the diagram,

(i) insert the asymptotes for the function f [2]

Now, $f(x) = \frac{3x-4}{x+4}$

Vertical asymptotes occur when the denominator equals 0.

When $x + 4 = 0$,

$$x = -4$$

$\therefore x = -4$ is a vertical asymptote of $f(x)$.

$$\lim_{x\to\infty} y = \lim_{x\to\infty} \frac{3x-4}{x+4}$$

$$= \lim_{x\to\infty} \frac{\frac{3x}{x} - \frac{4}{x}}{\frac{x}{x} + \frac{4}{x}}$$

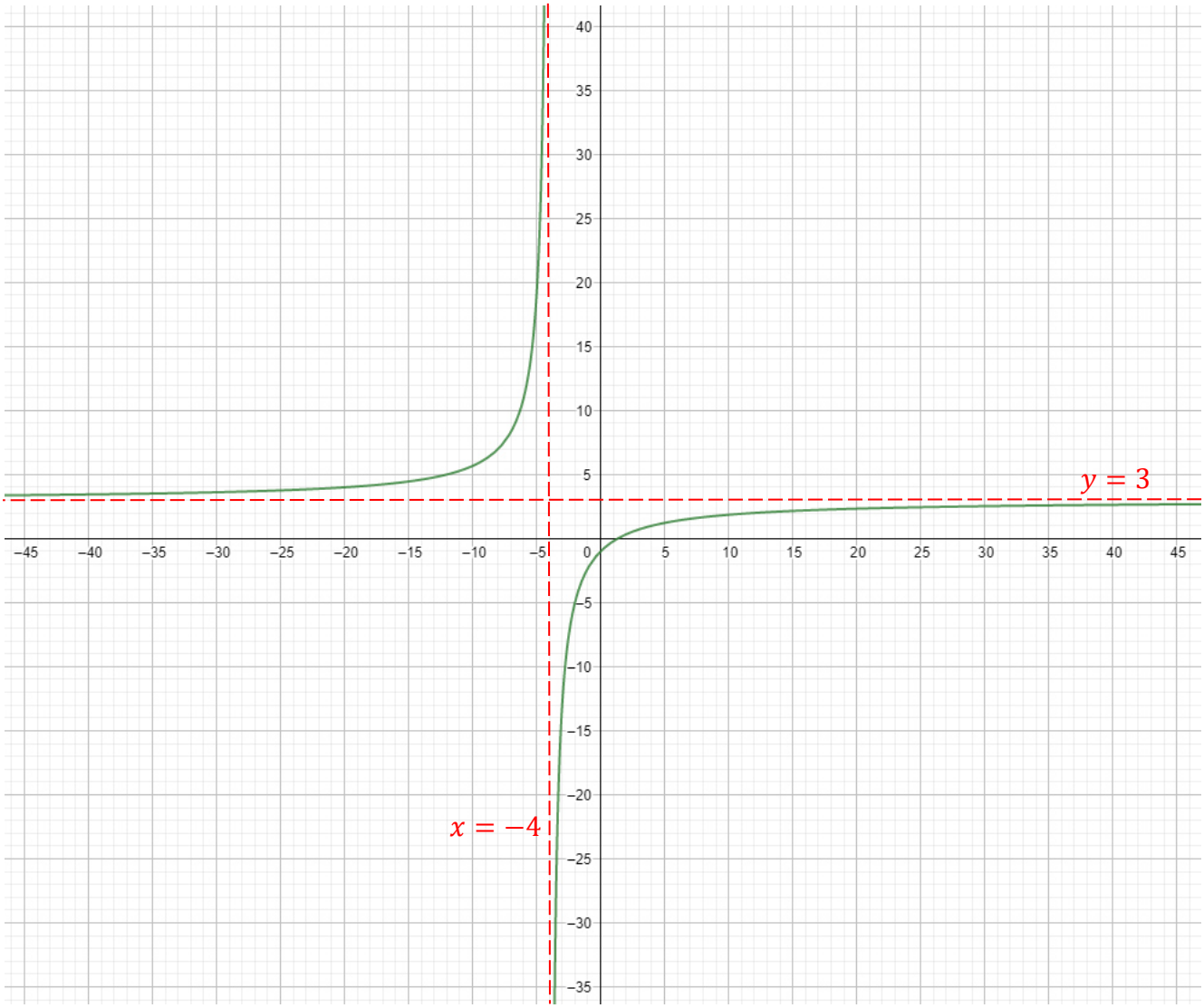
$$= \lim_{x\to\infty} \frac{3 - \frac{4}{x}}{1 + \frac{4}{x}}$$

$$= \frac{3}{1}$$

$$= 3$$

$\therefore y = 3$ is a horizontal asymptote.

The diagram is shown as follows:



(ii) sketch the graph of f^{-1}, the inverse of f showing the asymptotes for f^{-1} [4]

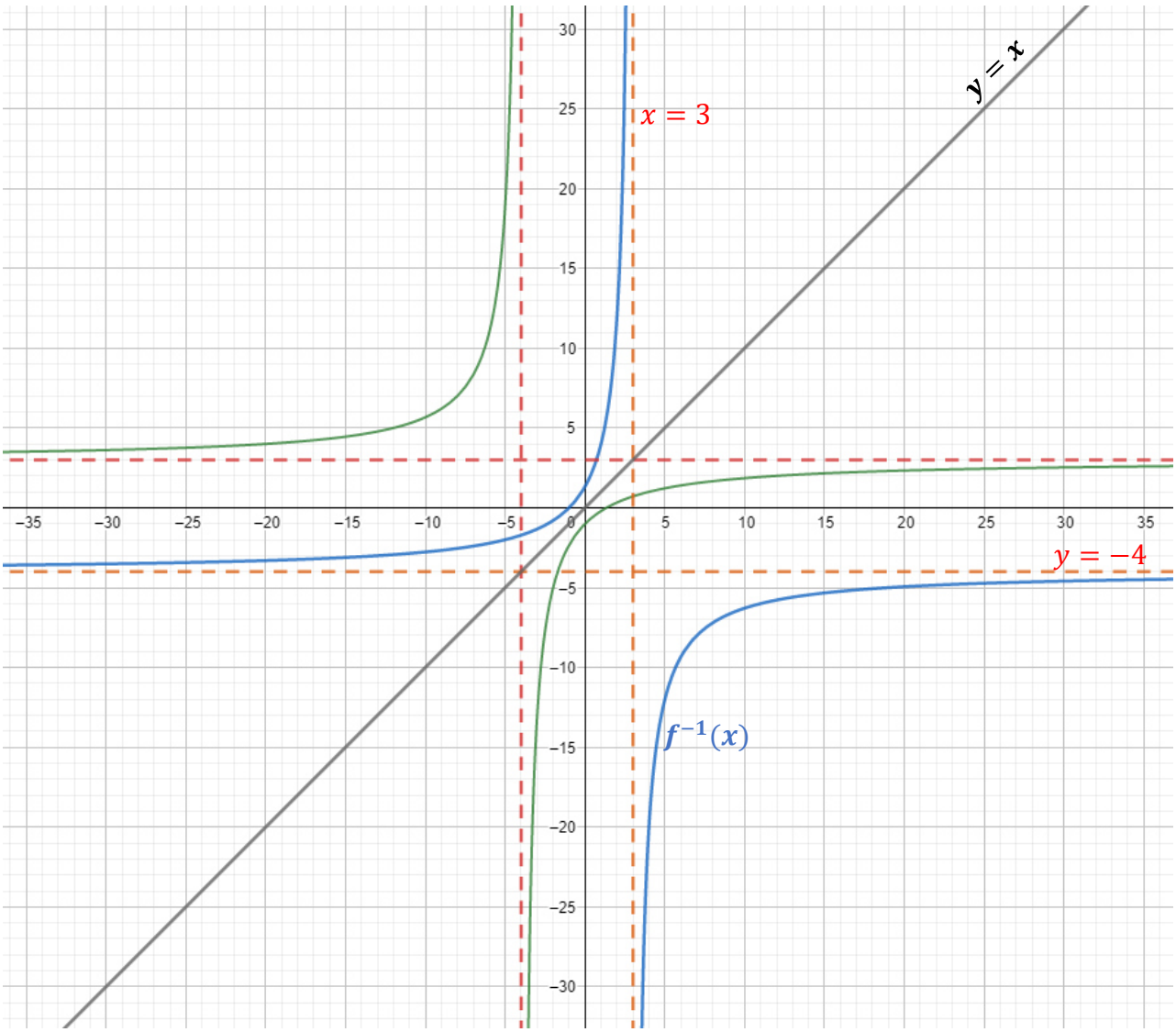
The graph of f^{-1} is a reflection of the graph of f in the line $y = x$.

For the graph of f^{-1},

the vertical asymptote is at $x = 3$

and the horizontal asymptote is at $y = -4$.

The diagram is shown as follows:



(c) Given that α, β and γ are the roots of the equation $x^3 + 3x + 2 = 0$, form an equation whose roots are $\beta\gamma, \alpha\gamma$ and $\alpha\beta$. [8]

The equation is:

$x^3 + 3x + 2 = 0$

where $a = 1$,

$b = 0$

$c = 3$

$d = 2$

Now,

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$= -\frac{0}{1}$$

$$= 0$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = \frac{c}{a}$$

$$= \frac{3}{1}$$

$$= 3$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

$$= -\frac{2}{1}$$

$$= -2$$

For the new roots, $\beta\gamma, \alpha\gamma$ and $\alpha\beta$:

$$\beta\gamma + \alpha\gamma + \alpha\beta = 3$$

$$(\beta\gamma)(\alpha\gamma) + (\beta\gamma)(\alpha\beta) + (\alpha\gamma)(\alpha\beta) = \alpha\beta\gamma^2 + \alpha\beta^2\gamma + \alpha^2\beta\gamma$$

$$= (\alpha\beta\gamma)(\gamma + \beta + \alpha)$$

$$= (-2)(0)$$

$$= 0$$

$$(\alpha\beta)(\alpha\gamma)(\beta\gamma) = \alpha^2\beta^2\gamma^2$$

$$= (\alpha\beta\gamma)^2$$

$$= (-2)^2$$

$$= 4$$

Hence, the new equation whose roots are $\beta\gamma, \alpha\gamma$ and $\alpha\beta$ is:

$$x^3 - (sum\ of\ roots)x^2 + (sum\ of\ the\ product\ of\ two\ of\ the\ roots\ at\ a\ time)x - (product\ of\ the\ roots) = 0$$

$$x^3 - 3x^2 + 0x - 4 = 0$$

$$x^3 - 3x^2 - 4 = 0$$

Total: 25 marks

June 2017 – Question 3 – Solution

(a) (i) Prove the identity

$$\tan(A+B) = \frac{\tan(A)+\tan(B)}{1-\tan(A)\tan(B)}$$ [4]

Taking L.H.S:

$$\tan(A+B) = \frac{\sin(A+B)}{\cos(A+B)}$$

$$= \frac{\sin A\cos B+\cos A\sin B}{\cos A\cos B-\sin A\sin B}$$

$$= \frac{\frac{\sin A\cos B}{\cos A\cos B}+\frac{\cos A\sin B}{\cos A\cos B}}{\frac{\cos A\cos B}{\cos A\cos B}-\frac{\sin A\sin B}{\cos A\cos B}}$$

$$= \frac{\frac{\sin A}{\cos A}+\frac{\sin B}{\cos B}}{1-\left(\frac{\sin A}{\cos A}\right)\left(\frac{\sin B}{\cos B}\right)}$$

$$= \frac{\tan(A)+\tan(B)}{1-\tan(A)\tan(B)}$$

$$= \text{R.H.S.}$$

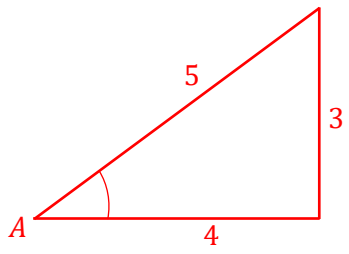
$$\therefore \tan(A+B) = \frac{\tan(A)+\tan(B)}{1-\tan(A)\tan(B)}$$

Q.E.D.

(ii) Given that $\sin A = \frac{3}{5}$ and $\cos B = \frac{-1}{2}$ where angle A is acute and angle B is obtuse, express $\tan(A+B)$ in the form $a + b\sqrt{3}$, where a and b are real numbers. [6]

We are given $\sin A = \frac{3}{5}$

$$= \frac{opp}{hyp}$$

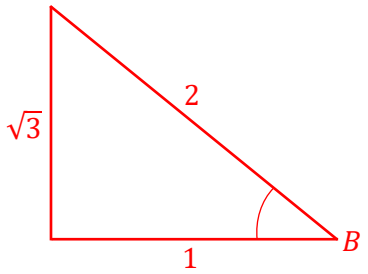


$\tan A = \frac{opp}{adj}$

$= \frac{3}{4}$

We are given $\cos B = \frac{-1}{2}$

$= \frac{adj}{hyp}$



Using Pythagoras' Theorem,

$a^2 + b^2 = c^2$

$1^2 + b^2 = 2^2$

$1 + b^2 = 4$

$b^2 = 4 - 1$

$b^2 = 3$

$b = \sqrt{3}$

So,

$$\tan B = \frac{opp}{adj}$$

$$= \frac{\sqrt{3}}{-1}$$

$$= -\sqrt{3}$$

Now,

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$= \frac{\frac{3}{4} + (-\sqrt{3})}{1 - \left(\frac{3}{4}\right)(-\sqrt{3})}$$

$$= \frac{\frac{3}{4} - \sqrt{3}}{1 + \frac{3\sqrt{3}}{4}}$$

$$= \left(\frac{3}{4} - \sqrt{3}\right) \div \left(1 + \frac{3\sqrt{3}}{4}\right)$$

$$= \frac{3-4\sqrt{3}}{4} \div \frac{4+3\sqrt{3}}{4}$$

$$= \frac{3-4\sqrt{3}}{4} \times \frac{4}{4+3\sqrt{3}}$$

$$= \frac{3-4\sqrt{3}}{4+3\sqrt{3}} \times \frac{4-3\sqrt{3}}{4-3\sqrt{3}}$$

$$= \frac{12-9\sqrt{3}-16\sqrt{3}+36}{16-27}$$

$$= \frac{48-25\sqrt{3}}{-11}$$

$$= -\frac{48}{11} + \frac{25}{11}\sqrt{3}$$

$\therefore \tan(A+B) = -\frac{48}{11} + \frac{25}{11}\sqrt{3}$ which is in the form $a + b\sqrt{3}$,

where $a = -\frac{48}{11}$ and $b = \frac{25}{11}$.

(b) Solve the equation $\sin^2\theta - 2\cos^2\theta + 3\cos\theta + 5 = 0$ for $0 \le \theta \le 4\pi$. [6]

$$\sin^2\theta - 2\cos^2\theta + 3\cos\theta + 5 = 0$$

$$(1 - \cos^2\theta) - 2\cos^2\theta + 3\cos\theta + 5 = 0$$

$$-3\cos^2\theta + 3\cos\theta + 6 = 0$$

$$\cos^2\theta - \cos\theta - 2 = 0$$

$$(\cos\theta + 1)(\cos\theta - 2) = 0$$

Either $\cos\theta + 1 = 0$ or $\cos\theta - 2 = 0$

$\cos\theta = -1$ $\cos\theta = 2$

$\theta = \pi, 3\pi$ [No solutions $\because -1 \le \cos\theta \le 1$]

$\therefore \theta = \pi, 3\pi$ for $0 \le \theta \le 4\pi$

(c) (i) Express $f(\theta) = 6\cos\theta + 8\sin\theta$ in the form $r\sin(\theta + \alpha)$ where $0 \le \alpha \le 90°$. [3]

Let $6\cos\theta + 8\sin\theta = r\sin(\theta + \alpha)$

$$8\sin\theta + 6\cos\theta = r\sin\theta\cos\alpha + r\cos\theta\sin\alpha$$

Equating coefficients gives:

$r\cos\alpha = 8$ $\rightarrow$ Equation 1

$r\sin\alpha = 6$ $\rightarrow$ Equation 2

Equation 2 $\div$ Equation 1 gives:

$$\frac{\sin\alpha}{\cos\alpha} = \frac{6}{8}$$

$$\tan\alpha = \frac{3}{4}$$

$$\alpha = \tan^{-1}\left(\frac{3}{4}\right)$$

$$\alpha = 36.9°$$

Now,

$$r = \sqrt{(8)^2 + (6)^2}$$

$$= \sqrt{64 + 36}$$

$$= \sqrt{100}$$

$$= 10$$

$$\therefore f(\theta) = 10\sin(\theta + 36.9°)$$

(ii) Hence, or otherwise, find the general solution of $f(\theta) = 2$. [6]

$f(\theta) = 10\sin(\theta + 36.9°)$ and $f(\theta) = 2$

Hence,

$$10\sin(\theta + 36.9°) = 2$$

$$\sin(\theta + 36.9°) = \frac{1}{5}$$

$$PV = 11.5°$$

$$\therefore \text{G.S. is } \theta + 36.9° = 180n + (-1)^n(11.5°) \qquad , n \in \mathbb{Z}$$

$$\theta = 180n + (-1)^n(11.5°) - 36.9° \qquad , n \in \mathbb{Z}$$

Total: 25 marks

June 2017 – Question 4 – Solution

(a) (i) The circle, C_1, with equation $x^2 + y^2 - 4x + 2y - 2 = 0$ and the circle C_2 have a common centre. Given that C_2 passes through the point $(-1, -2)$, express the equation of C_2 in the form $(x - h)^2 + (y - k)^2 = k$. [3]

$$x^2 + y^2 - 4x + 2y - 2 = 0$$

$$x^2 - 4x + y^2 + 2y = 2$$

$$x^2 - 4x + 4 + y^2 + 2y + 1 = 2 + 4 + 1$$

$$(x - 2)^2 + (y + 1)^2 = 7$$

which is in the form $(x - h)^2 + (y - k)^2 = k$,

where the centre is $(2, -1)$ and the radius, $r = \sqrt{7}$ units

Since C_1 and C_2 have a common centre, then centre of C_2 is $(2, -1)$.

Now, consider $(2, -1)$ and $(-1, -2)$.

$$\text{Radius} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(-1 - 2)^2 + \left(-2 - (-1)\right)^2}$$

$$= \sqrt{(-3)^2 + (-1)^2}$$

$$= \sqrt{9 + 1}$$

$$= \sqrt{10} \text{ units}$$

$\therefore$ The equation of C_2 is: $(x - 2)^2 + (y + 1)^2 = 10$.

(ii) The equation of the line L_1 is $x + 3y = 3$. Determine whether L_1 is a tangent to the circle, C_1, in (a)(i) on page 16. [7]

L_1: $x + 3y = 3$ → Equation 1

C_1: $x^2 + y^2 - 4x + 2y - 2 = 0$

$(x-2)^2 + (y+1)^2 = 7$ → Equation 2

Rearranging Equation 1 gives:

$x = 3 - 3y$ → Equation 3

Substituting Equation 3 into Equation 2 gives:

$$(3 - 3y - 2)^2 + (y+1)^2 = 7$$

$$(1 - 3y)^2 + (y+1)^2 - 7 = 0$$

$$1 - 6y + 9y^2 + y^2 + 2y + 1 - 7 = 0$$

$$10y^2 - 4y - 5 = 0$$

which is a quadratic where $a = 10, b = -4$ and $c = -5$.

Now,

$$b^2 - 4ac = (-4)^2 - 4(10)(-5)$$

$$= 16 + 200$$

$$= 216\ (> 0)$$

Since $b^2 - 4ac > 0$, then the line intersects the circle at two distinct points.

$\therefore L_1$ cannot be a tangent to the circle C_1.

(b) Let $\boldsymbol{P}(3,1,2)$ and $\boldsymbol{Q}(1,-2,4)$.

(i) Express the vector $\overrightarrow{\boldsymbol{PQ}}$ in the form $x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}} + z\hat{\boldsymbol{k}}$. [2]

Let $\boldsymbol{P}(3,1,2)$ and $\boldsymbol{Q}(1,-2,4)$.

So, $\overrightarrow{OP} = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$ and $\overrightarrow{OQ} = \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}$.

Now,

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$$

$$= \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix} - \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} 1-3 \\ -2-1 \\ 4-2 \end{pmatrix}$$

$$= \begin{pmatrix} -2 \\ -3 \\ 2 \end{pmatrix}$$

$= -2\hat{\boldsymbol{\imath}} - 3\hat{\boldsymbol{\jmath}} + 2\hat{\boldsymbol{k}}$ which is in the form $x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}} + z\hat{\boldsymbol{k}}$.

(ii) Determine the Cartesian equation of the plane which passes through the point $\boldsymbol{Q}$ and is perpendicular to $\overrightarrow{\boldsymbol{PQ}}$. [6]

Let $\boldsymbol{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, $\boldsymbol{n} = \begin{pmatrix} -2 \\ -3 \\ 2 \end{pmatrix}$ and $\boldsymbol{a} = \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}$.

The Cartesian equation of the plane is:

$$\boldsymbol{r} \cdot \boldsymbol{n} = \boldsymbol{a} \cdot \boldsymbol{n}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} -2 \\ -3 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ -3 \\ 2 \end{pmatrix}$$

$$-2x - 3y + 2z = (1)(-2) + (-2)(-3) + (4)(2)$$

$$-2x - 3y + 2z = -2 + 6 + 8$$

$$-2x - 3y + 2z = 12$$

(c) The vector equations of two lines, L_1 and L_2, are:

$$L_1 = -\hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}} - 2\hat{\boldsymbol{k}} + \alpha(-2\hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}} - 3\hat{\boldsymbol{k}})$$

$$L_2 = -2\hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}} - 4\hat{\boldsymbol{k}} + \beta(\hat{\boldsymbol{\imath}} - \hat{\boldsymbol{\jmath}} + \hat{\boldsymbol{k}})$$

(i) Show that L_1 and L_2 intersect. [5]

Required to show that L_1 and L_2 intersect.

$$L_1 = -\hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}} - 2\hat{\boldsymbol{k}} + \alpha(-2\hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}} - 3\hat{\boldsymbol{k}})$$

$$L_2 = -2\hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}} - 4\hat{\boldsymbol{k}} + \beta(\hat{\boldsymbol{\imath}} - \hat{\boldsymbol{\jmath}} + \hat{\boldsymbol{k}})$$

Equating coefficients gives:

$-1 - 2\alpha = -2 + \beta$ → Equation 1

$1 + \alpha = 1 - \beta$ → Equation 2

$-2 - 3\alpha = -4 + \beta$ → Equation 3

Equation 1 + Equation 2 gives:

$-\alpha = -1$

$\alpha = 1$

Substituting $\alpha = 1$ into Equation 2 gives:

$1 + 1 = 1 - \beta$

$2 = 1 - \beta$

$\beta = 1 - 2$

$\beta = -1$

Substituting $\alpha = 1$ and $\beta = -1$ into Equation 3 gives:

L.H.S $= -2 - 3(1) = -2 - 2 = -5$

R.H.S $= -4 + (-1) = -4 - 1 = -5$

Since L.H.S = R.H.S, then all three equations are satisfied by $\alpha = 1$ and $\beta = -1$.

$\therefore$ L_1 and L_2 intersect.

Q.E.D.

(ii) Hence, determine the coordinates of the point of intersection of the two lines. [2]

Required to determine the coordinates of the point of intersection of the two lines.

Substituting $\alpha = 1$ into L_1 gives:

$$-\hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}} - 2\hat{\boldsymbol{k}} + 1\left(-2\hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}} - 3\hat{\boldsymbol{k}}\right) = -\hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}} - 2\hat{\boldsymbol{k}} - 2\hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}} - 3\hat{\boldsymbol{k}}$$

$$= -3\hat{\boldsymbol{\imath}} + 2\hat{\boldsymbol{\jmath}} - 5\hat{\boldsymbol{k}}$$

Alternatively,

Substituting $\beta = -1$ into L_2 gives:

$$-2\hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}} - 4\hat{\boldsymbol{k}} + (-1)\left(\hat{\boldsymbol{\imath}} - \hat{\boldsymbol{\jmath}} + \hat{\boldsymbol{k}}\right) = -2\hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}} - 4\hat{\boldsymbol{k}} - \hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}} - \hat{\boldsymbol{k}}$$

$$= -3\hat{\boldsymbol{\imath}} + 2\hat{\boldsymbol{\jmath}} - 5\hat{\boldsymbol{k}}$$

$\therefore$ The point of intersection is $(-3, 2, -5)$.

Total: 25 marks

June 2017 – Question 5 – Solution

(a) Determine the value of k for which

$$f(x) = \begin{cases} \frac{x^5-1}{x-1}, & x \neq 1 \\ k, & x = 1 \end{cases}$$

is continuous for all values of x. [4]

Firstly, $f(1) = k$.

Now,

$$\lim_{x\to 1} f(x) = \lim_{x\to 1} \frac{x^5-1}{x-1}$$

$$= \lim_{x\to 1} \frac{(x-1)(x^4+x^3+x^2+x+1)}{(x-1)}$$

$$= \lim_{x\to 1} (x^4 + x^3 + x^2 + x + 1)$$

$$= (1)^4 + (1)^3 + (1)^2 + (1) + 1$$

$$= 1 + 1 + 1 + 1 + 1$$

$$= 5$$

For $f(x)$ to be continuous for all values of x, $\lim_{x\to 1} f(x) = f(1)$.

$\therefore k = 5$

(b) A curve, C, is described parametrically by the equations $x = 5t + 3$ and

$y = t^3 - t^2 + 2$.

(i) Find $\frac{dy}{dx}$ in terms of t. [3]

$x = 5t + 3$

$\frac{dx}{dt} = 5$

And,

$y = t^3 - t^2 + 2$

$\frac{dy}{dt} = 3t^2 - 2t$

Using the Chain Rule,

$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$

$= 3t^2 - 2t \times \frac{1}{5}$

$= \frac{1}{5}(3t^2 - 2t)$

(ii) Hence, determine all points of C such that $\frac{dy}{dx} = 0$. [6]

When $\frac{dy}{dx} = 0$,

$\frac{1}{5}(3t^2 - 2t) = 0$

$3t^2 - 2t = 0$

$t(3t - 2) = 0$

Either $t = 0$ or $3t - 2 = 0$

$t = \frac{2}{3}$

When $t = 0$,

$x = 5(0) + 3$

$= 0 + 3$

$= 3$

And,

$y = (0)^3 - (0)^2 + 2$

$= 0 - 0 + 2$

$= 2$

Hence, the point is $(3, 2)$.

When $t = \frac{2}{3}$,

$x = 5\left(\frac{2}{3}\right) + 3$

$= \frac{10}{3} + 3$

$= \frac{19}{3}$

And,

$y = \left(\frac{2}{3}\right)^3 - \left(\frac{2}{3}\right)^2 + 2$

$= \frac{8}{27} - \frac{4}{9} + 2$

$= \frac{50}{27}$

Hence, the point is $\left(\frac{19}{3}, \frac{50}{27}\right)$.

$\therefore$ All points of C such that $\frac{dy}{dx} = 0$ are $(3, 2)$ and $\left(\frac{19}{3}, \frac{50}{27}\right)$.

(c) (i) Given that $y = \sqrt{2 + 2x^2}$, show that

(a) $y\frac{dy}{dx} - 2x = 0$

$y = \sqrt{2 + 2x^2}$

$y = (2 + 2x^2)^{\frac{1}{2}}$

$\frac{dy}{dx} = \frac{1}{2}(4x)(2 + 2x^2)^{\frac{1}{2}-1}$

$\frac{dy}{dx} = 2x(2 + 2x^2)^{-\frac{1}{2}}$

Now, taking L.H.S,

$y\frac{dy}{dx} - 2x = (2 + 2x^2)^{\frac{1}{2}}(2x)(2 + 2x^2)^{-\frac{1}{2}} - 2x$

$= 2x - 2x$

$= 0$

$= \text{R.H.S.}$

$\therefore y\frac{dy}{dx} - 2x = 0$

Q.E.D.

(b) $\frac{d^2y}{dx^2} - \frac{4}{y^3} = 0$ [9]

$\frac{dy}{dx} = 2x(2 + 2x^2)^{-\frac{1}{2}}$

Let $u = 2x$ and $v = (2+2x^2)^{-\frac{1}{2}}$

$\frac{du}{dx} = 2$ $\frac{dv}{dx} = \left(-\frac{1}{2}\right)(4x)(2+2x^2)^{-\frac{3}{2}}$

$\frac{dv}{dx} = -2x(2+2x^2)^{-\frac{3}{2}}$

So,

$$\frac{d^2y}{dx^2} = u\frac{dv}{dx} + v\frac{du}{dx}$$

$$= 2x(-2x)(2+2x^2)^{-\frac{3}{2}} + (2+2x^2)^{-\frac{1}{2}}(2)$$

$$= -4x^2(2+2x^2)^{-\frac{3}{2}} + 2(2+2x^2)^{-\frac{1}{2}}$$

Now, taking L.H.S,

$$\frac{d^2y}{dx^2} - \frac{4}{y^3} = -4x^2(2+2x^2)^{-\frac{3}{2}} + 2(2+2x^2)^{-\frac{1}{2}} - \frac{4}{\left[(2+2x^2)^{\frac{1}{2}}\right]^3}$$

$$= -4x^2(2+2x^2)^{-\frac{3}{2}} + 2(2+2x^2)^{-\frac{1}{2}} - 4(2+2x^2)^{-\frac{3}{2}}$$

$$= 2(2+2x^2)^{-\frac{3}{2}}[-2x^2 + (2+2x^2)^1 - 2]$$

$$= 2(2+2x^2)^{-\frac{3}{2}}(-2x^2 + 2 + 2x^2 - 2)$$

$$= 2(2+2x^2)^{-\frac{3}{2}}(0)$$

$$= 0$$

$$= \text{R.H.S.}$$

$$\therefore \frac{d^2y}{dx^2} - \frac{4}{y^3} = 0$$

Q.E.D.

(ii) Hence, find the value of $\frac{d^2y}{dx^2}$ when $x = 0$. [3]

$$\frac{d^2y}{dx^2} = -4x^2(2+2x^2)^{-\frac{3}{2}} + 2(2+2x^2)^{-\frac{1}{2}}$$

When $x = 0$,

$$\frac{d^2y}{dx^2} = -4(0)^2(2+2(0)^2)^{-\frac{3}{2}} + 2(2+2(0)^2)^{-\frac{1}{2}}$$

$$= 2(2)^{-\frac{1}{2}}$$

$$= (2)^{1-\frac{1}{2}}$$

$$= (2)^{\frac{1}{2}}$$

$$= \sqrt{2}$$

$\therefore \frac{d^2y}{dx^2} = \sqrt{2}$ when $x = 0$.

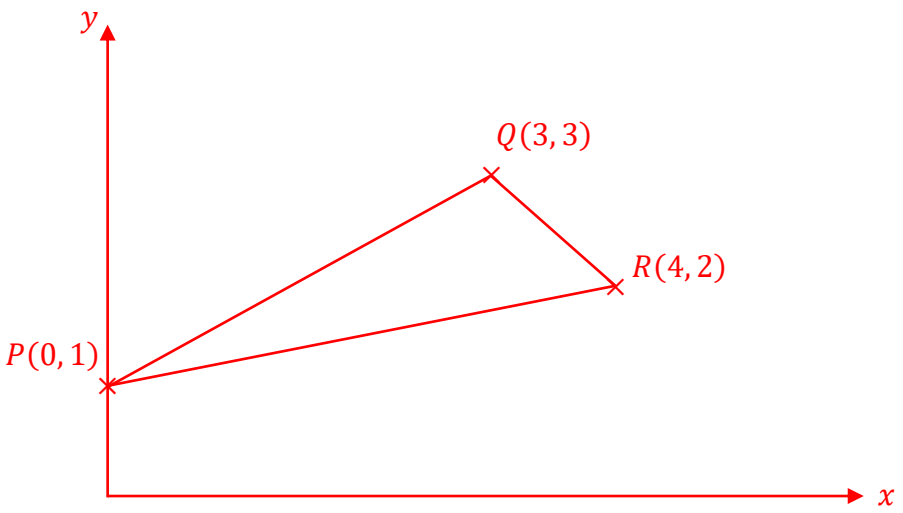
Total: 25 marks

June 2017 – Question 6 – Solution

(a) Triangle PQR has vertices $P(0,1)$, $Q(3,3)$ and $R(4,2)$.

(i) On the axes below, sketch triangle PQR. [1]

Triangle PQR has vertices $P(0,1)$, $Q(3,3)$ and $R(4,2)$.



(ii) Determine the equations of EACH of the following:

- PQ
- QR
- PR [7]

Consider $P(0,1)$ and $Q(3,3)$:

Gradient $= \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{3-1}{3-0}$

$= \frac{2}{3}$

The y-intercept occurs at $(0,1)$.

Hence, the equation of PQ is: $y = \frac{2}{3}x + 1$

Consider $Q(3,3)$ and $R(4,2)$:

Gradient $= \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{2-3}{4-3}$

$= -1$

Substituting $m = -1$ and point $(3,3)$ into $y - y_1 = m(x - x_1)$ gives:

$y - 3 = -1(x - 3)$

$y - 3 = -x + 3$

$y = -x + 3 + 3$

$y = -x + 6$

Hence, the equation of QR is: $y = -x + 6$

Consider $P(0,1)$ and $R(4,2)$:

Gradient $= \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{2-1}{4-0}$

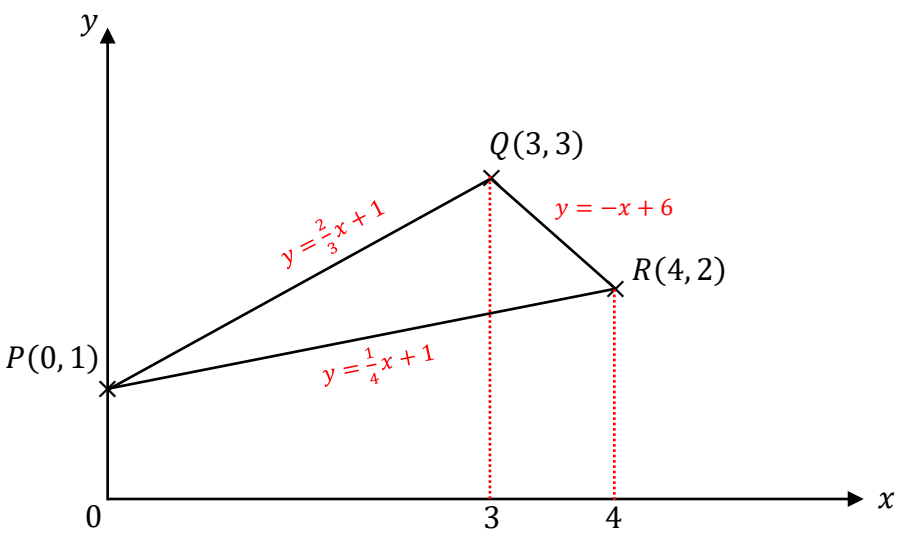
$= \frac{1}{4}$

The y-intercept occurs at $(0,1)$.

Hence, the equation of PR is: $y = \frac{1}{4}x + 1$

(iii) Hence, use integration to determine the area of triangle PQR. [7]

Required to use integration to determine the area of triangle PQR.



Now,

Area of triangle = Area under PQ + Area under QR − Area under PR

Area under $PQ = \int_a^b y\,dx$

$$= \int_0^3 \left(\frac{2}{3}x + 1\right) dx$$

$$= \left[\frac{2}{3}\left(\frac{x^2}{2}\right) + x\right]_0^3$$

$$= \left[\frac{x^2}{3} + x\right]_0^3$$

$$= \left(\frac{3^2}{3} + 3\right) - \left(\frac{0^2}{3} + 0\right)$$

$$= \frac{9}{3} + 3 - 0$$

$$= 6 \text{ units}^2$$

Area under $QR = \int_a^b y\,dx$

$$= \int_3^4 (-x + 6)\,dx$$

$$= \left[-\frac{x^2}{2} + 6x\right]_3^4$$

$$= \left[-\frac{(4)^2}{2} + 6(4)\right] - \left[-\frac{(3)^2}{2} + 6(3)\right]$$

$$= \left(-\frac{16}{2} + 24\right) - \left(-\frac{9}{2} + 18\right)$$

$$= -8 + 24 + \frac{9}{2} - 18$$

$$= \frac{5}{2} \text{ units}^2$$

Area under $PR = \int_a^b y\,dx$

$$= \int_0^4 \left(\frac{1}{4}x + 1\right) dx$$

$$= \left[\frac{1}{4}\frac{x^2}{2} + x\right]_0^4$$

$$= \left[\frac{x^2}{8} + x\right]_0^4$$

$$= \left(\frac{4^2}{8} + 4\right) - \left(\frac{0^2}{8} + 0\right)$$

$$= \frac{16}{8} + 4 - 0$$

$$= 2 + 4 - 0$$

$$= 6 \text{ units}^2$$

Hence,

Area of triangle = Area under PQ + Area under QR − Area under PR

$$= 6 + \frac{5}{2} - 6$$

$$= \frac{5}{2} \text{ units}^2$$

∴ The area of the triangle is $\frac{5}{2}$ units^2.

(b) The voltage in a circuit, V, satisfies the equation $\frac{dV}{dt} + \frac{V}{2.5} = 0$. Given that $V = 25$ volts when $t = 0$ seconds, write an expression for V in terms of t. [5]

$$\frac{dV}{dt} + \frac{V}{2.5} = 0$$

$$\frac{dV}{dt} = \frac{-V}{\frac{5}{2}}$$

$$\frac{dV}{dt} = \left(-\frac{2}{5}\right) V$$

Separating variables:

$$\frac{1}{V}\, dV = -\frac{2}{5}\, dt$$

Integrating both sides:

$$\int \frac{1}{V}\, dV = \int -\frac{2}{5}\, dt$$

$$\ln V = -\frac{2}{5}t + c$$

We are given that $V = 25$ volts when $t = 0$ seconds.

$$\ln 25 = -\frac{2}{5}(0) + c$$

$$c = \ln 25$$

Now,

$$\ln V = -\frac{2}{5}t + \ln 25$$

$$V = e^{-\frac{2}{5}t + \ln 25}$$

$$V = e^{-\frac{2}{5}t} \cdot e^{\ln 25}$$

$$V = 25e^{-\frac{2}{5}t}$$

$\therefore$ An expression for V in terms of t is: $V = 25e^{-\frac{2}{5}t}$

(c) Given that $\int_{-1}^{3}[3f(x)+g(x)]\,dx = 5$ and $\int_{-1}^{3}[5f(x)-2g(x)]\,dx = 1$, determine

- $\int_{-1}^{3} f(x)\,dx$
- $\int_{-1}^{3} g(x)\,dx$ [5]

We are given that,

$$\int_{-1}^{3}[3f(x)+g(x)]\,dx = 5$$

$$3\int_{-1}^{3} f(x)\,dx + \int_{-1}^{3} g(x)\,dx = 5$$

And,

$$\int_{-1}^{3}[5f(x)-2g(x)]\,dx = 1$$

$$5\int_{-1}^{3} f(x)\,dx - 2\int_{-1}^{3} g(x)\,dx = 1$$

Let $m = \int_{-1}^{3} f(x)\,dx$ and $n = \int_{-1}^{3} g(x)\,dx$.

We have,

$3m + n = 5 \quad \rightarrow$ Equation 1

$5m - 2n = 1 \quad \rightarrow$ Equation 2

Multiplying Equation 1 by 2 gives:

$6m + 2n = 10 \quad \rightarrow$ Equation 3

Equation 2 + Equation 3 gives:

$11m = 11$

$m = \frac{11}{11}$

$m = 1$

Substituting $m = 1$ into Equation 2 gives:

$5(1) - 2n = 1$

$5 - 2n = 1$

$2n = 5 - 1$

$2n = 4$

$n = \frac{4}{2}$

$n = 2$

$\therefore \int_{-1}^{3} f(x)\, dx = 1$ and $\int_{-1}^{3} g(x)\, dx = 2$

Total: 25 marks